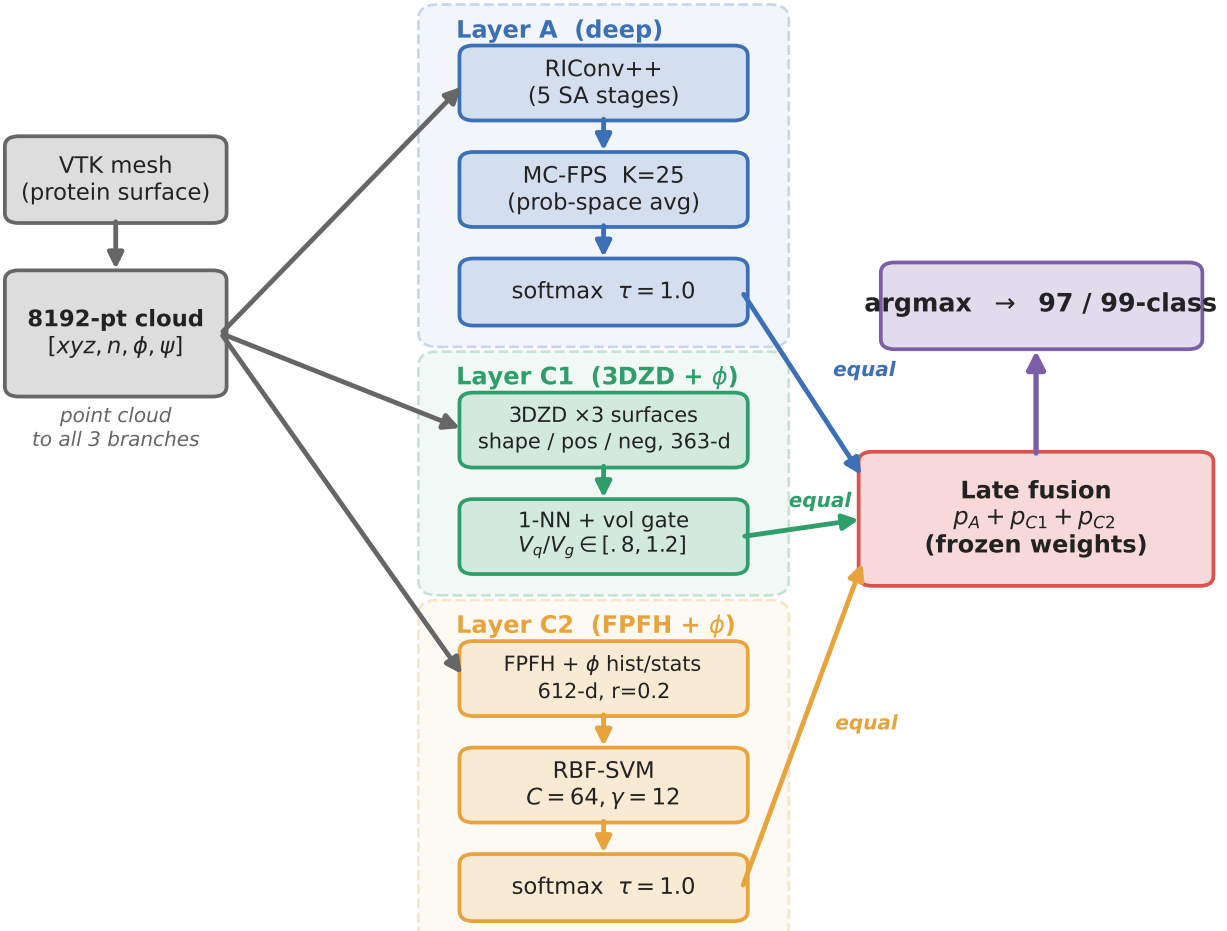
VTK mesh
(protein surface)
8192-pt cloud
[xyz, n, ϕ, ψ]
point cloud
to all 3 branches
Layer A (deep)
RIConv++
(5 SA stages)
MC-FPS K=25
(prob-space avg)
softmax τ = 1.0
Layer C1 (3DZD + ϕ)
3DZD ×3 surfaces
shape / pos / neg, 363-d
1-NN + vol gate
V_q/V_g ∈ [.8, 1.2]
Layer C2 (FPFH + ϕ)
FPFH + ϕ hist/stats
612-d, r=0.2
RBF-SVM
C = 64, γ = 12
softmax τ = 1.0
equal
equal
equal
Late fusion
p_A + p_C1 + p_C2
(frozen weights)
argmax → 97 / 99-class